25900669 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course(VAC) - MG3VACTXLA01 - CHEMICAL PROCESSING IN TEXTILES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Under which classification does flame-resistant finish come? [U] / [CO1]

   a). Basic/routine finish b). Functional finish

   c). Aesthetic Finishes d). Colouration finish

2. What is a disadvantage of stock dyeing? [U] / [CO3]

   a). Difficult shade matching b). Expensive and time consuming

   c). Cannot dye cotton d). Produces dull shades

3. What type of dye is commonly used for dyeing wool and silk fibers? [K] / [CO3]

   a). Vat dyes b). Acid dyes

   c). Sulfur dyes d). Disperse dyes

4. In direct printing, how is the design applied to the fabric? [U] / [CO4]

   a). By removing colour from fabric b). By applying printing paste directly on fabric

   c). By coating fabric with resin d). By dyeing the entire fabric

5. What component is not required in the jet printing process? [U] / [CO4]

a). Screens b). Blocks

c). Rollers d). All of the above

6. What is the primary benefit of using nanotechnology in textile finishing? [U] / [CO2]

a). Improved Durability b). Enhanced water repellency

c). Increased UV protection d). All of the above

7. In flat-bed screen printing, the number of screens corresponds to: [U] / [CO4]

a). Number of prints b). Number of colors

c). Type of fabric d). Complexity of the squeegee

8. What happens if fabric is over bleached? [K] / [CO2]

a). Improved strength b). Increased weight

c). Fiber damage and yellowing d). No change

9. Which of the following finishes is considered temporary and removed after washing? [K] / [CO2]

a). Resin finish b). Weighting with metallic salts

c). simple calendering d). Flame retardant finish

10. What is the purpose of reducing agents in vat dyeing? [U] / [CO3]

a). To improve in color fastness b). To increase dye solubility

c). To reduce the dye to its leuco form d). To fix the dye on the fabric

[1x10 = 10]

Part B

Fill in the Blanks Questions

Answer all questions

Each question carries **1** marks

11. Gigging was traditionally done using brushes made of _____ plants. [U] / [CO2]

12. Solution dyeing involves adding colorants to the _____solution before fibre production. [U] / [CO3]

13. Basic or routine finishes are also known as_____finishes. [K] / [CO1]

14. At the manufacturing level, moth-proofing can involve modifying the sulphur-containing _____ in wool. [K] / [CO2]

15. In product dyeing garments are dyed after _____. [U] / [CO3]

16. To ensure proper alignment when stamping, many blocks have _____ carved into them. [U] / [CO4]

17. Acid dyes are suitable for dyeing _____fibers such as nylon. [U] / [CO3]

18. Stencil printing influenced the later development of _____ printing. [U] / [CO4]

19. Discharge printing is most effective on fabrics dyed with _____ dyes. [U] / [CO4]

20. Soil and stain-resistant finishes reduce the need for frequent washing or heavy _____. [U] / [CO2]

[1x10 = 10]

Part C

Answer in One Word or One Sentence

Answer all questions

Each question carries **1** marks

21. Which natural method helps reduce microbes on fabrics at home? [K] / [CO2]

22. What controls the design in digital textile printing? [U] / [CO4]

23. What is union dyeing? [U] / [CO3]

24. What is the common application of plasma technology? [K] / [CO2]

25. Under which classification does sueding come? [K] / [CO1]

26. Which technique is used to deposit fibers in flocking besides mechanical methods? [K] / [CO2]

27. What is dope dyeing? [U] / [CO3]

28. What is a mordant? [U] / [CO4]

29. What is a characteristic of reactive dyes? [K] / [CO3]

30. Name one other application for sublimation methods besides textiles. [U] / [CO4]

[1x10 = 10]

Part D

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

31. In functional finishes, explain how water-repellent finishes improve the comfort and utility of fabrics. [K] / [CO1]

32. Why is shrinkage resistant finishing important for wool fabrics? [U] / [CO2]

33. What is an advantage of yarn dyeing? [K] / [CO3]

34. Why is sublimation printing not suitable for untreated cotton? [U] / [CO4]

35. Explain the embossing process in textile finishing? [U] / [CO2]

36. What is the main advantage of jet dyeing? [U] / [CO3]

[2x5 = 10]

Part E

Long Answer Questions

Answer any **2** questions

Each question carries **5** marks

37. Explain the process and importance of durable press finishes in textiles. [U] / [CO2]

38. Explain aesthetic finishes and their two types? [U] / [CO2]

39. What is dip dyeing, and why are fabrics dipped repeatedly? [U] / [CO3]

40. Explain the roller printing process step by step, including the main machinery and components involved. [U] / [CO4]

[5x2 = 10]